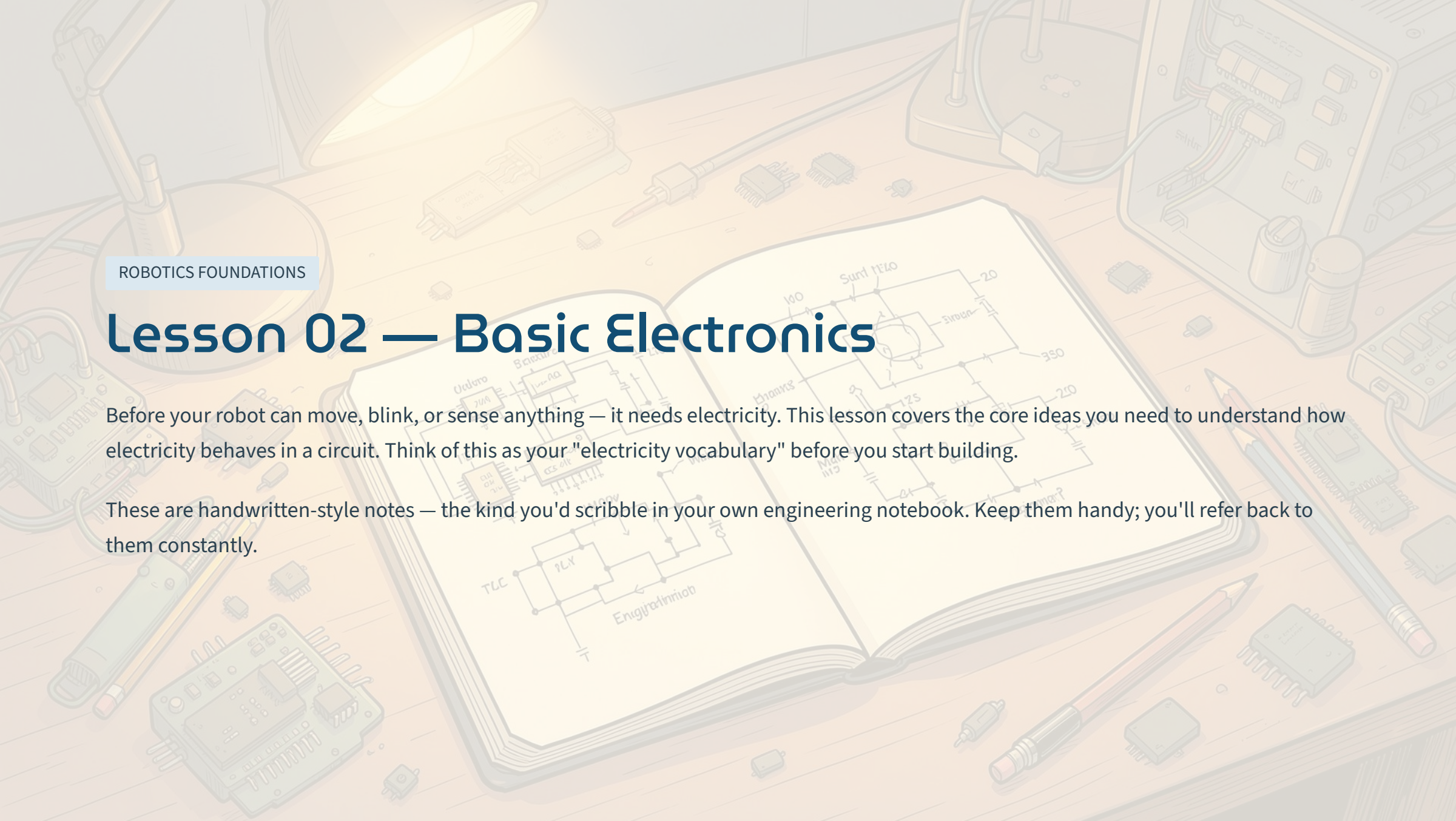


The background is a detailed illustration of a workspace. In the center is an open notebook with two pages of handwritten-style circuit diagrams. The left page shows a circuit with components labeled 'TLC', '12V', and 'Engin...'. The right page shows a more complex circuit with labels like 'Surf PEZO', '20', '350', '200', and '24'. Surrounding the notebook are various electronic components: integrated circuits, resistors, capacitors, and a breadboard. A desk lamp is in the top left, and a soldering iron is in the bottom right. The overall color palette is warm and slightly desaturated, with a focus on the notebook and its contents.

ROBOTICS FOUNDATIONS

Lesson 02 — Basic Electronics

Before your robot can move, blink, or sense anything — it needs electricity. This lesson covers the core ideas you need to understand how electricity behaves in a circuit. Think of this as your "electricity vocabulary" before you start building.

These are handwritten-style notes — the kind you'd scribble in your own engineering notebook. Keep them handy; you'll refer back to them constantly.

Voltage, Current & Power

Three big ideas form the foundation of all electronics. Here's the water-flow analogy that makes them click — imagine electricity flowing through a wire like water flowing through a pipe.

Voltage (V)

Unit: Volts (V)

Electrical "pressure" that pushes current through a circuit. Like water pressure in a pipe — higher voltage means more push.

Common values: AA battery = 1.5V, USB = 5V, Car battery = 12V

Current (I)

Unit: Amperes (A)

The flow rate of electric charge. Like the amount of water flowing past a point each second.

Small currents are measured in milliamps (mA): 1 A = 1000 mA

Power (P)

Unit: Watts (W)

How fast energy is used or delivered. A higher-wattage component consumes more energy per second.

Formula: $P = V \times I$



YouTube

Voltage, Current, and Power Explained – Laying the Foundation for Ba...

While voltage, current, and power will soon be second nature to you, before we get into circuit analysis, let's take a moment to explain the relationship between...



- ❏ **Energy** is power used over time. A 10W LED running for 2 hours uses 20 Watt-hours of energy. Your robot's battery stores a finite amount of energy — once it's gone, the robot stops!

The Water-Flow Analogy

This analogy is your best friend when learning electronics. Every time you're confused about voltage or current, picture water in pipes.



Water Pressure = Voltage

A tall water tank creates high pressure at the bottom — just like a high-voltage battery creates strong electrical push.



Water Flow = Current

A wide pipe lets more water flow. A thick wire lets more current flow. Narrow pipe = less flow = higher resistance.



Pump = Battery

The battery is the pump that keeps the "water" (current) moving around the loop.

Key Relationship

Power tells you how much work a component does per second.

$$P = V \times I$$

If a motor runs at **12V** and draws **2A**:

$$P = 12 \times 2 = 24 \text{ W}$$

That motor uses 24 Joules of energy every second it runs.

Resistors — What Are They?

A **resistor** is a component that opposes the flow of electric current. It "resists" current — the higher the resistance, the less current flows for a given voltage. Resistors are everywhere in electronics.

[Resistors Explained & Ohm's Law Made Simple! | What Are Resistors? & How Do Resistors Work? - YouTube](#)

Why Do We Use Resistors?

→ Limit Current

Protect sensitive components (like LEDs) from too much current.

→ Create Voltage Drops

Divide voltage between parts of a circuit.

→ Dissipate Heat

Convert excess electrical energy into heat safely.

How Resistance Affects Current

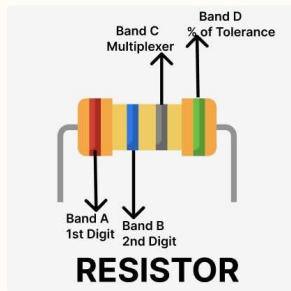
More resistance → less current. Less resistance → more current.

This is the heart of Ohm's Law (next section!).



Resistors get warm when current flows through them — that's energy being lost as heat. This is normal, but high-power resistors need space to cool!

Types of Resistors



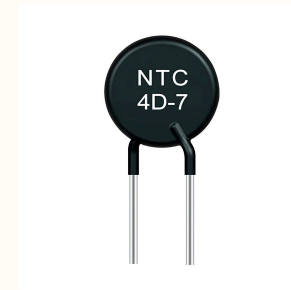
Fixed Resistor

Has one set resistance value (e.g., 220Ω , $1k\Omega$). Identified by colored bands. Most common type in beginner circuits.



Potentiometer

A **variable resistor** — you can adjust resistance by turning a knob or sliding a control. Used for volume controls, speed controls, etc.



Thermistor

A resistor whose value **changes with temperature**. Used in temperature-sensing circuits. Resistance drops as it gets hotter (usually).

Circuit Symbol: A zigzag line (US) or rectangle (international). The symbol appears in every circuit diagram you'll draw.

Ohm's Law

Ohms Law Explained - The basics circuit theory - YouTube

Ohm's Law is the most important equation in basic electronics. It connects voltage, current, and resistance in one simple relationship.

$$V = I \times R$$

Three Forms — Same Equation

Find Voltage

$$V = I \times R$$

Find Current

$$I = \frac{V}{R}$$

Find Resistance

$$R = \frac{V}{I}$$

Quick Example

A 9V battery is connected to a 100Ω resistor. How much current flows?

$$I = \frac{V}{R} = \frac{9}{100} = 0.09 \text{ A} = 90 \text{ mA}$$

90 milliamps of current flows through the resistor.

Directly Proportional

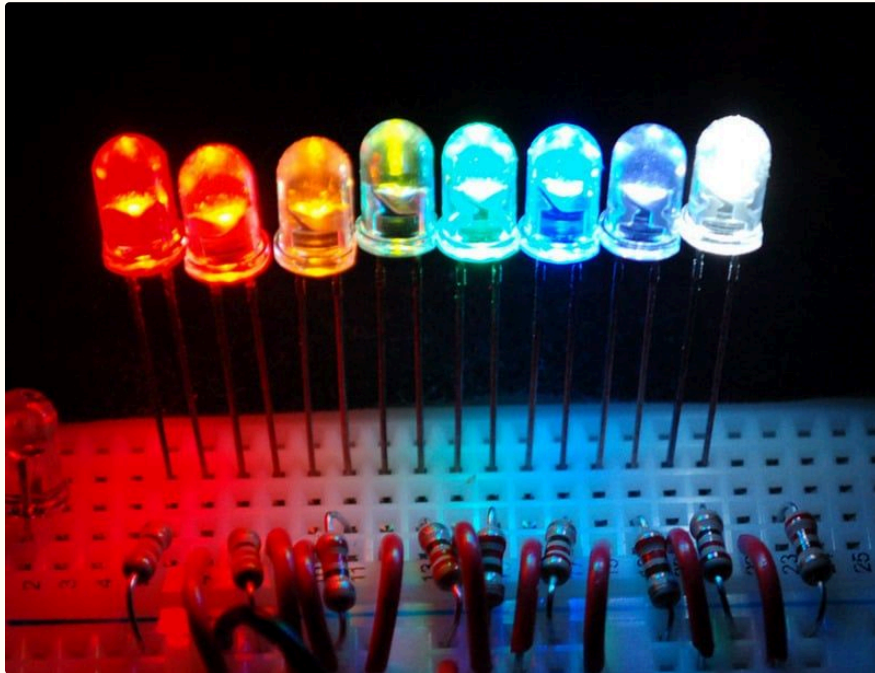
Voltage ↑ → **Current** ↑ (if R stays the same). Double the voltage, double the current.

Inversely Proportional

Resistance ↑ → **Current** ↓ (if V stays the same). Double the resistance, halve the current.

LEDs — Light Emitting Diodes

[How LED Works - Unravel the Mysteries of How LEDs Work! - YouTube](#)



An **LED (Light Emitting Diode)** is a special component that emits light when current flows through it. LEDs are efficient, long-lasting, and come in many colors.

How It Works

LEDs are **diodes** — they only allow current to flow in **one direction**. Connect it backwards and it won't light up!

Anode (+)

The **longer leg**. Connects to positive voltage. Current enters here.

Cathode (-)

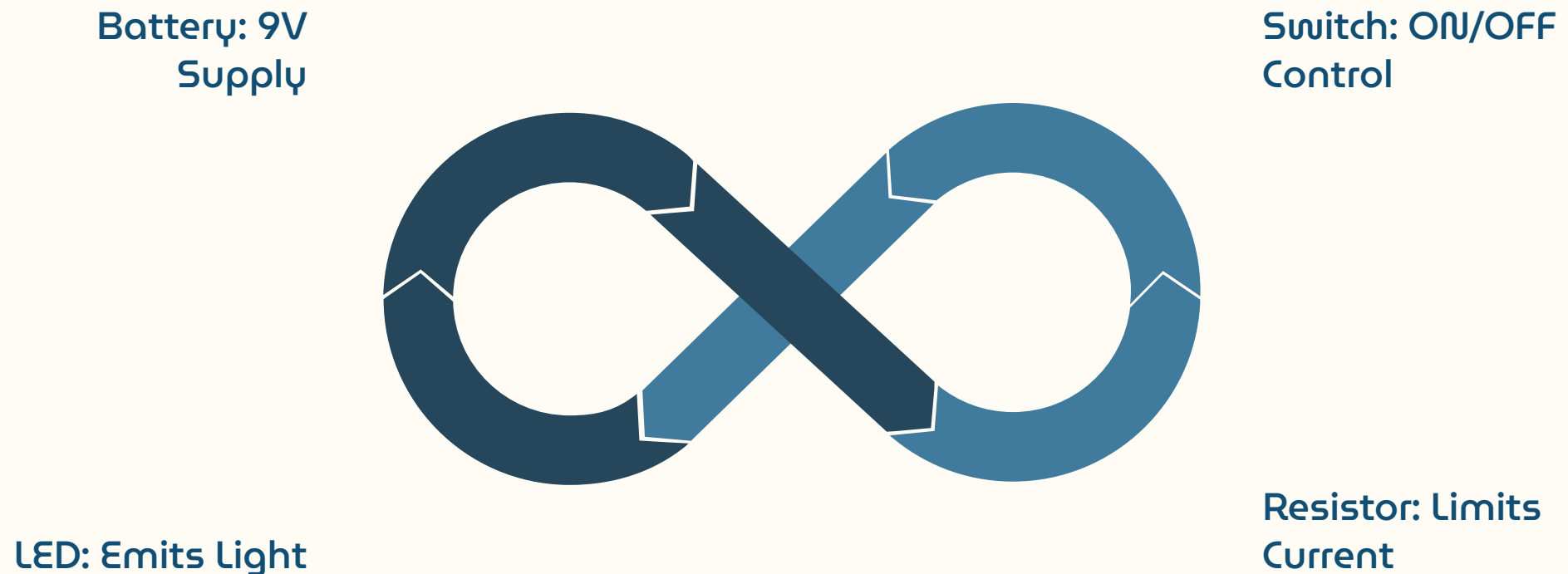
The **shorter leg**. Connects to negative/ground. Current exits here. Often has a flat edge on the LED body.

- ❏ LEDs need a **current-limiting resistor** in series — without it, too much current flows and the LED burns out instantly!

LED + Switch + Battery Circuit

[How to Connect Multiple LED's with 9V Battery, Switch in Series Connection | Simple Electric Circuit - YouTube](#)

Let's put it all together. This is the simplest complete circuit — a battery, a switch, a resistor, and an LED. When the switch is closed, current flows and the LED lights up.



Current flows from the battery's positive terminal, through the switch, through the resistor, through the LED (anode to cathode), and back to the battery's negative terminal — but *only* when the switch is **ON**.

🔌 Switch ON

Circuit is **complete**. Current flows. LED lights up. Resistor protects the LED from excess current.

🔌 Switch OFF

Circuit is **broken**. No current flows. LED is dark. Battery is not drained.

Reading Circuit Diagrams

Engineers use standard symbols to draw circuits. Learn these four symbols and you can read almost any beginner schematic.



Battery

Two parallel lines — long line is + (positive), short line is – (negative). Provides voltage to the circuit.



LED

A diode triangle with two arrows pointing away (light emission). Flat side = cathode (–).



Resistor

A zigzag line. Opposes current flow. Value is written next to it (e.g., 220 Ω).



Switch

A line with a gap and a lever. Open = OFF (no current). Closed = ON (current flows).

Remember — The Important Formulas

Keep these three equations in your head (or on a sticky note on your desk). They are the foundation of everything you'll do in electronics and robotics.



Ohm's Law

$$V = I \times R$$

Voltage = Current × Resistance



Power Formula

$$P = V \times I$$

Power = Voltage × Current



Rearranged Forms

$$I = \frac{V}{R} \quad R = \frac{V}{I}$$

Use whichever form matches what you need to find.

Quick Rules to Remember

- Voltage pushes, current flows, resistance opposes
- LEDs need a resistor — always!
- Current flows from + to – (conventional flow)
- Switch OFF = broken circuit = no current
- More resistance = less current

What's Next?

Now that you understand voltage, current, resistance, and LEDs — you're ready to start **building real circuits** on a breadboard. Next up: series and parallel circuits, and how to choose the right resistor for your LED!